

1. Prepare each file and add explicit links

VCF inputs (four arms)

1: Five genome slices:

NB72462M,
NG131FQA1I, 0GOOR_HG002,
HG004_GRCh38, HG005_GRCh38
(all .vcf.gz)

2: 104 single-sample cohort VCFs

3: Complete HG005_GRCh38 VCF

4: Complete NB72462M VCF

+ ClinVar; + frequencies (Arm 2)

Then normalize (GRCh38).

VCF-RDFizer conversion

VCF → RDF graph per file.

Link plug-in: same VCF

Rules: linker.ttl

Pinned GRCh38 references

SPDI from position + alleles

Gene interval overlap

→ separate .links.nt.

RDF graphs + their link triples

Shared variant identity:

genome call → SPDI ← ClinVar call
frequency-panel call → the same SPDI
Edges: vcfl:sameVariantAs

Gene links for policy selection:

genome call → Ensembl gene identifier
Edge: vcfl:overlapsGene

Shared identifiers let queries join the files.

2. Build a validated release view

Where the policy is defined

Case JSON + gene list

→ generate.py → policy.ttl

ODRL file consents and withdrawal;

28 cancer genes: clinical care only.

Arm 4: + panel, region, variant.

Purposes use DUO terms.

Policy plug-in: evaluate

Select file records and gene overlaps.

Match requester and purpose.

A prohibition overrides permission.

No permission → withhold.

Write permitted data and links.

Request context

--assignee: requester identifier

--purpose: DUO purpose term

CC, DS, and GRU requesters,
plus the participant's physician
in Arm 4.

participant RDF + links

release view

Policy plug-in: check

Compare with a VCF-derived reference.

Check ownership and dangling links.

Pass → validated view.nt.gz

One release view per requester.

shared annotation RDF + links

3. Query each release view

SPARQL query enters here

carriers.rq: 81 ACMG genes;
ALT carrier; PASS/missing FILTER;
ClinVar review; join calls by SPDI.
rare.rq: add frequency < 1%.

QLever indexes and queries

Validated participant release view

+ ClinVar RDF and links

+ frequency RDF and links (Arm 2).

Run the same query for each requester.

Returned information

File, sample, gene, chromosome,
position, REF/ALT, SPDI,
ClinVar classification, genotype;
and frequency for rare.rq.